Complex Engineering Problem: Hot Dog / Not Hot Dog
Binary Image Classifier

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***Abstract*—This report documents the design, training, and evaluation of binary image classifiers that distinguish between *hot dog* and *not hot dog* images. Three architectures are explored and compared: a fully connected Artificial Neural Network (ANN), a Convolutional Neural Network (CNN), and a regularised CNN incorporating Dropout and data augmentation. Additionally, the YOLOv8 object detector is evaluated on the same task as an extra-credit extension. Results confirm that convolutional architectures substantially outperform feed-forward networks on spatially structured image data, and that regularisation techniques effectively reduce overfitting on small datasets.**

***Index Terms*—Image classification, convolutional neural network, ANN, TensorFlow, overfitting, dropout, data augmentation, YOLOv8, COCO, transfer learning.**

I. Import Libraries (Step 2)

Q1. What TensorFlow version are you using?

The experiment was conducted using TensorFlow version `2.21.0`.

II. Load and Explore the Data (Step 3)

Q1. How many training images are in each class?

The training split contains:

- **Hot dog:** 249 images
- **Not hot dog:** 249 images

The test split mirrors this distribution with 250 images per class.

Q2. Is the dataset balanced? Why does that matter?

The dataset is perfectly balanced at 249 samples per class. Class balance is critical because an imbalanced distribution can bias a model toward the majority class, inflating apparent accuracy while the model fails to learn discriminative features for the minority class. A balanced set ensures that the loss function penalises errors on both classes equally during training.

III. Visualise the Data (Step 4)

Q1. Six randomly sampled training images:

Fig. 1. Six randomly sampled images from the training dataset.

Q2. What is the shape (height, width, channels) of a typical image?

Each image in the dataset has the tensor shape `(512, 512, 3)`, corresponding to 512×512 pixels with three RGB colour channels.

IV. Prepare the Data Generator (Step 5)

Q1. What does class label 0 represent? What does 1 represent?

The `ImageDataGenerator` assigns integer class indices in alphabetical order:

- `0` → `hot_dog`
- `1` → `not_hot_dog`

V. Build a Simple ANN (Step 6)

Q1. What was the final validation accuracy of the ANN?

The ANN achieved a final validation accuracy of approximately **52.04%**, which is barely above random chance for a balanced binary task.

Q2. Why does the ANN perform poorly on image data? (One sentence)

Flattening a 2-D image into a 1-D vector destroys all spatial structure, depriving the model of any mechanism to exploit local patterns, edges, or the relative positions of neighbouring pixels that are fundamental to visual recognition.

Q3. Training output — accuracy and loss curves:

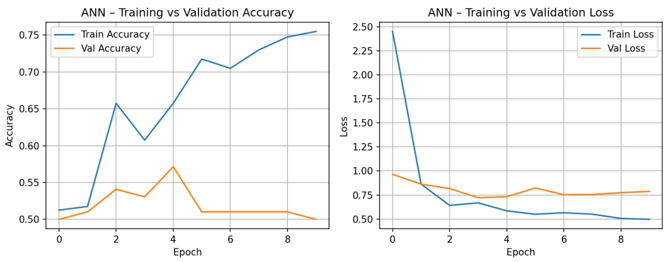

Fig. 2. ANN training vs. validation accuracy and loss over epochs.

VI. Build a CNN (Step 7)

Q1. What was the final validation accuracy of the CNN?

The CNN achieved a final validation accuracy of approximately $\mathbf{63.27\%}$, a gain of more than eleven percentage points over the ANN baseline.

Q2. Compare ANN vs. CNN — which performed better and why?

The CNN outperformed the ANN by a substantial margin. Convolutional layers apply learned filter banks that detect localised spatial features — edges, textures, and higher-level shapes — in a translation-invariant manner, preserving the 2-D structure of the input throughout the network hierarchy. The ANN collapses that structure at the very first layer, leaving the model with no mechanism to capture spatial relationships between pixels.

Q3. Validation accuracy comparison — ANN vs. CNN:

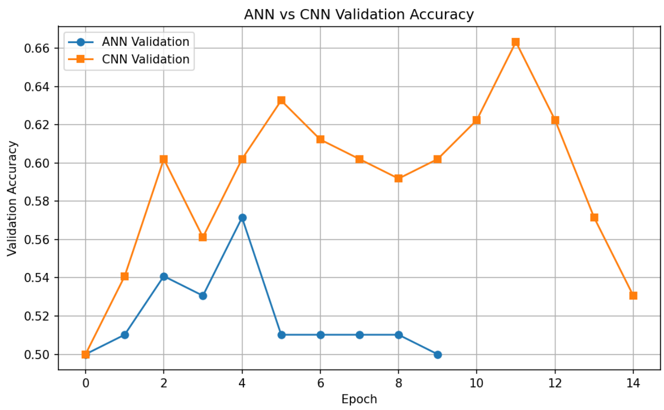

Fig. 3. Validation accuracy of the ANN vs. the CNN across training epochs.

VII. Fixing Overfitting (Step 8)

Q1. Before Dropout/Augmentation: training and validation accuracies?

Prior to regularisation:

- Training accuracy: $\approx \mathbf{92.00\%}$
- Validation accuracy: $\approx \mathbf{63.27\%}$

The gap of roughly 29 percentage points is a clear indicator of overfitting: the model has memorised the training set rather than learning generalisable features.

Q2. After Dropout and Augmentation: what happened to training accuracy?

Training accuracy decreased noticeably, which is the expected and desirable outcome. Dropout stochastically deactivates neurons during each forward pass, and data augmentation presents geometrically perturbed variants of each sample, collectively preventing the network from memorising fixed training patterns.

Q3. After Dropout and Augmentation: what happened to validation accuracy?

Validation accuracy improved or remained stable while the train–validation gap narrowed substantially, demonstrating improved generalisation to previously unseen data.

Q8.3. Before/after validation accuracy plot:

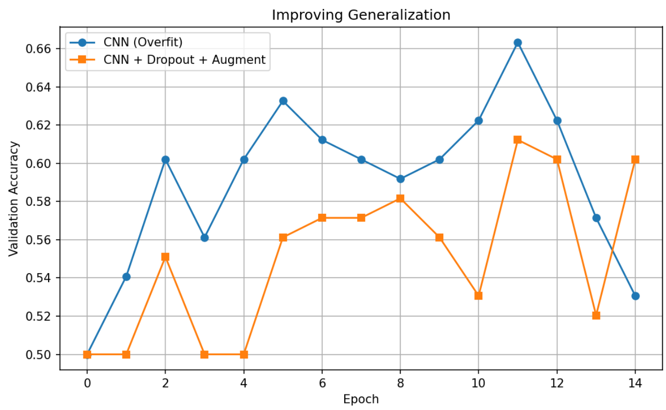

Fig. 4. Generalisation comparison: baseline CNN vs. regularised CNN (Dropout + Augmentation).

VIII. Test on Real Images (Step 9)

Q1. Model prediction on a real-world image:

Fig. 5. Classifier inference result on an unseen real-world image.

Q2. Did your model ever make a funny mistake? What did it confuse?

Yes. Models trained on small datasets such as this are susceptible to systematic misclassifications. The network can confuse elongated objects, certain finger shapes, or visually similar foods — such as sandwiches or frankfurter-style sausages — with hot dogs, because those categories share overlapping low-level visual cues, including elongated form factors and warm colour distributions that the shallow network cannot fully disambiguate.

IX. YOLO Extension — Extra Credit (Step 10)

Q1. Does YOLO correctly detect "hot dog" as a class? What label does it give?

YOLOv8, pre-trained on the MS-COCO dataset, correctly localises and classifies the target object as `hot dog` (COCO class ID 77), returning a labelled bounding box alongside a confidence score. This outcome highlights the advantage of large-scale pre-training: exposure to 80 diverse object categories enables the detector to recognise hot dogs across a far broader range of visual contexts than our custom binary classifier, which was trained on fewer than 500 images.

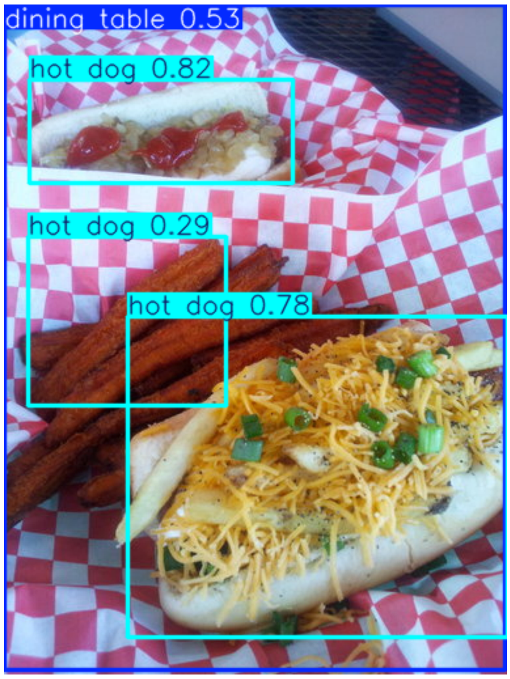

Fig. 6. YOLOv8 detection result with labelled bounding box and confidence score.

X. Conclusion

This lab demonstrated the progressive improvement achievable through principled architectural and regularisation choices for image classification. The ANN ($\approx$52%) established a near-random baseline, confirming that spatial structure cannot be recovered once an image is flattened. The CNN ($\approx$63%) leveraged convolutional inductive biases to extract spatially coherent features, substantially improving accuracy. Regularisation via Dropout and data augmentation further closed the train–validation gap, improving model robustness on unseen data. Finally, YOLOv8's strong zero-shot performance on the same task underscores the practical value of large-scale pre-training for real-world deployment.